

Lesson 23: Matrix Operations and Products

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginner (1)
Understand Dimensions	Correctly determines if multiplication is possible based on dimensions (m x n and n x p) and correctly identifies the dimensions of the result (m x p).	Identifies if multiplication is possible but makes a minor error in determining the resulting dimension.	Understands the need for matching internal dimensions but cannot determine the final dimension.	Does not understand that dimensions must match.
Calculation Accuracy	All scalar multiplications and additions for each element are accurate. Result is correct.	One or two minor calculation errors (e.g., arithmetic, sign error) that do not indicate a lack of understanding.	Several arithmetic errors, indicating a need for practice, but the process is mostly correct.	Fundamental misunderstanding of the matrix multiplication process.

Process & Procedure	Correctly applies the row-by-column rule for every element in the resulting matrix.	Applies the row-by-column rule correctly but skips steps in intermediate work.	Shows some steps but struggles to apply the row-by-column rule consistently.	No clear process or strategy used.
Organization & Presentation	Final answer is clearly stated in matrix notation with legible, organized intermediate work.	Final answer is in matrix notation, but intermediate work is slightly disorganized.	Final answer is present but messy; difficult to follow steps.	Work is disorganized, sloppy, or missing.